

Code and Data
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Overview:

All raw data of the experiments in the main text of the paper are in the file “experimental_data.zip”, except for those of new Table 3, trials 10~23 of Table 4 (the old Table 3 in previous versions of the manuscript), and the experiments in Appendix B.

All the software code of the Python programs for generating the above data are located in the “programs” directory.

For the code, data, and instructions for Figure 9, new Table 3, Table B.1 and trials 10~23 of Table 4, see:

https://github.com/gphehub/pnn2204/tree/main/additional_experiments_mar2026/readme.pdf

About the programs:

The MNIST data set and *mnist_loader.py* (not included here) will be called by all programs except *weight_extractor.py*. Available at:

<https://github.com/mnielsen/neural-networks-and-deep-learning/archive/master.zip>

All programs with the names ending with “S” are for the networks using sigmoid function.

- *main2204_2Para6S.py*:

A batch processing file that completes all the training and analysis (except the function of *weight_extractor.py*) of a network.

It includes *main2204_2Para_load_6S.py*.

It will call *network2Para6S.py* and *network2Para_load_6S.py*.

It will produce the following files:

biases&weights_initialized.pickle

biases&weights_optimized.pickle

network2Para6S.pkl

classification_accuracy.csv

disagreed_results.csv

Main parameters that can be changed:

[784,48,35,10],[784,50,10]: the size of the networks.

epochs in the function *SGD*: the number of epochs that the networks are trained in total.

epochs_sep in the function *SGD*: the number of epochs that the networks are trained separately.

- *main2204_2Para_load_6S.py*:
A batch processing file that analyzes a trained network.
It will call *network2Para_load_6S.py*.
It will read *biases&weights_optimized.pickle*, and produce *disagreed_results.csv*.
- *network2Para6S.py*:
The program that trains a network.
It will produce the following files:
biases&weights_initialized.pickle
biases&weights_optimized.pickle
network2Para6S.pkl
classification_accuracy.csv
- *network2Para_load_6S.py*:
The program that analyzes a trained network.
It will read *biases&weights_optimized.pickle*, and produce *disagreed_results.csv*.

All programs with the names ending with “R” are for the networks using ReLU function.

All programs with the names ending with “T” are for the networks using Tanh function.

- *weight_extractor.py*:
The program that extracts the weights of the outputs of the neurons of the last hidden layer when calculating the weighted inputs to the output neurons.
It will read *biases&weights_optimized.pickle* and produce *weights_at_the_output_layer.csv*.

Experimental data file description:

- *biases&weights_initialized.pickle*: The initial biases and weights of the network.
- *biases&weights_optimized.pickle*: The optimized biases and weights of the network.
- *classification_accuracy.csv*: The classification accuracies at each epochs and their maximums.
Notations (left: symbols in the file, right: symbols in the paper):
NN1: α_1
NN1p: α'_1
NN2: α_2
NN2p: α'_2
Para: α_{para}
- *disagreed_results.csv*: The analysis result of a trained network.
Notations (left: symbols/phrases in the file, right: symbols/phrases in the paper):

NN1: α_1

NN1p: α'_1

NN2: α_2

NN2p: α'_2

Para: α_{para}

Results when both subnetworks are right: Type I results

Results when only network1 is wrong: Type II results

Results when only network2 is wrong: Type III results

Results when both subnetworks are wrong but the combined network is right: Type IV results

- *network2Para6S.pkl* / *network2Para6R.pkl* / *network2Para6T.pkl*: The network at the end of the training.
- *weights_at_the_output_layer.csv*: The weights of the outputs of the neurons of the last hidden layer when calculating the weighted inputs to the output neurons.